$$\frac{6y}{6} = \frac{60}{6} \quad \text{[on both sides, divide by 6]}$$

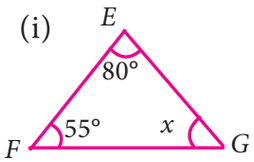
$$y = 10°$$

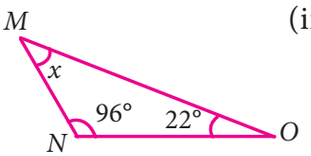
$\angle UWY = 6y + 2 = 6(10) + 2 = 62°$.

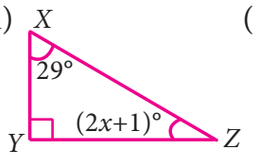
We can conclude that $\angle XWV = \angle UWY$, because they are vertically opposite angles as well as exterior angles.

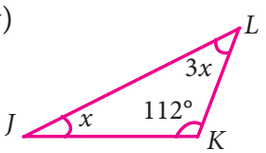
Exercise 4.1

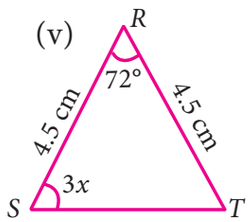
1. Can 30°, 60° and 90° be the angles of a triangle?
2. Can you draw a triangle with 25°, 65° and 80° as angles?
3. In each of the following triangles, find the value of x.

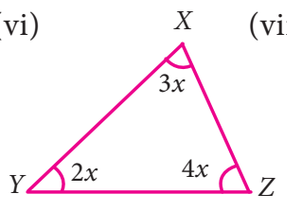
(i) 

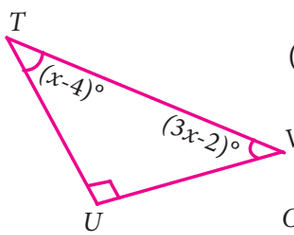
(ii) 

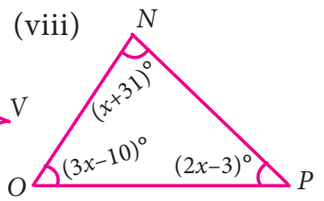
(iii) 

(iv) 

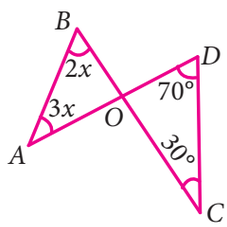
(v) 

(vi) 

(vii) 

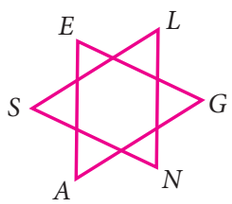
(viii) 

4. Two line segments $\overline{AD}$ and $\overline{BC}$ intersect at O. Joining $\overline{AB}$ and $\overline{DC}$ we get two triangles, $\triangle AOB$ and $\triangle DOC$ as shown in the figure. Find the $\angle A$ and $\angle B$.



5. Observe the figure and find the value of

$\angle A + \angle N + \angle G + \angle L + \angle E + \angle S$.



6. If the three angles of a triangle are in the ratio 3 : 5 : 4, then find them.
7. In $\triangle RST$, $\angle S$ is 10° greater than $\angle R$ and $\angle T$ is 5° less than $\angle S$, find the three angles of the triangle.
8. In $\triangle ABC$, if $\angle B$ is 3 times $\angle A$ and $\angle C$ is 2 times $\angle A$, then find the angles.
9. In $\triangle XYZ$, if $\angle X : \angle Z$ is 5 : 4 and $\angle Y = 72°$. Find $\angle X$ and $\angle Z$

7th_Maths_T2_EM_Chp4.indd 73 29-05-2019 12:19:35